



Ecological change increases malaria risk in the Brazilian Amazon

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Ecological change in the Brazilian Amazon is closely linked to human mobility and health. Mining, agriculture, logging, and other activities alter highly diverse ecological and demographic contexts and subsequent exposure to diseases such as malaria. Studies that have attempted to quantify the impact of deforestation on malaria in the Brazilian Amazon have produced conflicting results. However, they varied in methodology and data sources. Most importantly, all studies used annual data, neglecting the subannual seasonal dynamics of malaria. Here, we fill the knowledge gap on the subannual relationship between ecological change in the Brazilian Amazon and malaria transmission. Using the highest spatiotemporal resolution available, we estimated the effect of deforestation on malaria cases between 2003 and 2022 using a stratified Bayesian spatiotemporal hierarchical zero-inflated Poisson model fitted with the Integrated Nested Laplace Approximation. The model was also stratified by state. We found that a 1% increase in 1-mo lagged deforestation increased malaria cases in a given month and municipality by 6.3% [95% credible interval (CrI): 6.2, 6.5%]. Based on an interaction term included in the model, the effect of deforestation on malaria was even larger in areas with higher forest cover. We found that the coefficients for deforestation and mobility were highly variable when stratified by state. Our results provide detailed evidence that, on average, deforestation increases malaria transmission, but that the relationship is not spatiotemporally uniform. These results have implications for stratifying malaria control interventions based on ecological dynamics to help Brazil achieve its goal of malaria elimination by 2035.

malaria | deforestation | Amazon

Ecological change in the Brazilian Amazon is closely linked to human mobility and health (1–5). Historically, the most dramatic ecological changes have been marked by economic booms and busts associated with the exploitation of natural resources (6). In the early 1970s, a major project of national integration promoted by the military government encouraged migration to the Amazon. Agricultural settlement programs were initiated, especially in the 1980s (7). Population growth was rapid due to migration, but most migrants had little or no knowledge and previous exposure to malaria (7). Deforestation was intense.

The process of deforestation can alter the abundance and quality of mosquito breeding habitats, depending on the characteristics of the mosquitoes and on how it affects the local ecosystem (8, 9). The primary malaria vector in Brazil, *Nyssorhynchus* (*Ny.*, formerly *Anopheles*) *darlingi* (10–14), prefers breeding habitats with clean, shallow, deep, partially sunlit and clear water with a pH close to neutral, with aquatic vegetation but free of decomposing organic matter (15). Within the undisturbed forest, where light is scarce, the ideal conditions for *Ny. darlingi* breeding are rarely found. However, at the forest fringe (the boundary between the forest and the deforested area), ideal breeding conditions exist, and mosquito density and human-mosquito contact tend to be higher (8, 16, 17). As a result, malaria transmission has become a public health concern, with a sixfold increase in malaria cases between 1975 and 1990, reaching nearly 600,000 cases (7, 18).

Several policies have been implemented to mitigate deforestation and malaria transmission (19). The National Malaria Control Program (NMCP) was implemented in 2003, resulting in a steady decline in cases. In 2016, there were approximately 143,000 malaria cases, the lowest in 37 y (18). In 2006, Brazil tightened regulations on deforestation, leading to the most significant and sustained decline in deforestation rates in history. However, deforestation showed an upward trend in 2016 and escalated to very high levels starting in 2019. In 2021, the highest annual deforestation rate since 2006 was observed (<http://www.obt.inpe.br/prodes/>). This increase was fueled by changes in legislation introduced by former President Bolsonaro, which contributed to the expansion of illegal mining (mainly artisanal mining, called *garimpo*) (20). Considering the period from 1985 to 2022, 40.7% of all *garimpo* in the Amazon and 62.3% of all *garimpo* in indigenous lands

Significance

Deforestation in the Amazon is thought to increase malaria. However, some studies have found the opposite or no effect. Limitations of these studies include the use of annual data (ignoring transmission seasonality), coarse spatial scale, limited time series, and often neglect frontier malaria dynamics. We use the most detailed data available to quantify the monthly effect of deforestation on malaria, controlling for socioeconomic and environmental variables, including human mobility. We find that a 1% increase in deforestation is associated with a 6.31% rise in malaria cases. This study highlights state-level heterogeneity in the relationship between deforestation and malaria, and underscores the importance of accounting for ecological context and mobility, providing a nuanced understanding of how deforestation affects malaria risk.

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in the Amazon were opened from 2018 and 2022 (21). This has resulted in major health challenges (22, 23). While the number of malaria cases decreased by 36% between 2017 and 2022, malaria cases originating in indigenous and *garimpo* areas increased by 62% and 257%, respectively, during the same period (23).

This relationship between environmental change and malaria transmission in the Brazilian Amazon is central to the concept of frontier malaria, which operates across multiple spatial and temporal scales. It is characterized by a rapid increase in malaria cases following initial deforestation, and the consequent increased vector density, and more frequent human-vector contact. Frontier malaria includes a temporal component, with high transmission in the years following ecological changes, and a progressive decline over time (7, 14, 24). Evidence has accumulated since its initial conceptualization to support and extend this concept (14, 25–27).

Several studies have attempted to quantify the effect of deforestation on malaria transmission in the Brazilian Amazon due to the link between ecological change and the disease. The mechanistic relationship between the two is through the dynamics of frontier malaria. It is important to avoid biased language and stick to precise word choice when discussing this topic. These studies have varied in their methodological approach, data sources, spatial and temporal scales for both malaria and deforestation data, metrics for deforestation, assumptions of the lagged effect of deforestation, and the introduction of the concept of frontier malaria into the analysis (7, 14). As a result, 32% of studies found a negative association, 26% found a null or ambiguous association, and only 11% found a positive association between deforestation and malaria transmission (28). Annual data have been used to analyze the relationship between deforestation and malaria for periods ranging from one to thirteen years (29–31). However, the concept of frontier malaria and the entomological literature indicate that the ecology of *Ny. darlingi* responds to deforestation on subannual time scales (25). It is important to note that different definitions of deforestation imply nuanced interpretations inherent to each metric (28). For instance, the deforestation rate (percentage of forest area lost per year) differs from the absolute area of forest cleared at each time point, particularly when comparing regions with different land areas. Some studies have found a linear and

positive correlation (of varying magnitude) between deforestation and malaria (29, 31), while others have found evidence of a quadratic relationship (32). The impact of deforestation on malaria transmission was found to be nonhomogenous across space. It may vary according to the active/passive frontier (29) and the degree of previous degradation (30). One study suggested that forest fragmentation/high forest edge density was the main driver of malaria transmission in the Amazon, while others found no relationship between forest cover and malaria transmission (28).

It is crucial to understand the relationship between deforestation and malaria, given the continued ecological changes in the Amazon and Brazil's new elimination plan launched in 2022. The plan has four phases: i) reducing cases to less than 68,000 by 2025, ii) eliminating *Plasmodium falciparum* cases and deaths by 2030, iii) achieving zero cases and deaths by 2035, and iv) preventing the reintroduction of malaria from 2035 (33).

This study utilizes the longest time series (2003–2022) of data available with the finest spatial (municipality) and temporal (monthly) resolution to characterize and quantify the impact of deforestation on malaria in the Brazilian Amazon. The models employed in this study control for other environmental and social variables, as well as the mobility of malaria-infected individuals. To examine the correlation between monthly deforestation rates and malaria cases at the municipal level in the Brazilian Amazon, we utilized a Bayesian spatiotemporal hierarchical zero-inflated Poisson model. The model was stratified by the eight states that comprise the region.

Results

We used land cover/use data from MapBiomas (34) and malaria data from the Brazilian Ministry of Health (*Materials and Methods*). From 2003 to 2022, 5,426,227 cases of malaria were reported, and 207,287 km² of forest cover was removed in the Brazilian Amazon (4.13% of the total area) (Fig. 1 *A* and *B*). By law, officially demarcated indigenous lands and conservation units should be preserved (Fig. 1 *C*). Recent evidence shows that demarcation not only protects these areas, but also results in higher secondary forest growth (35). However, illegal deforestation has occurred in most of these areas between 2003 and 2022 (Fig. 1 *D*).

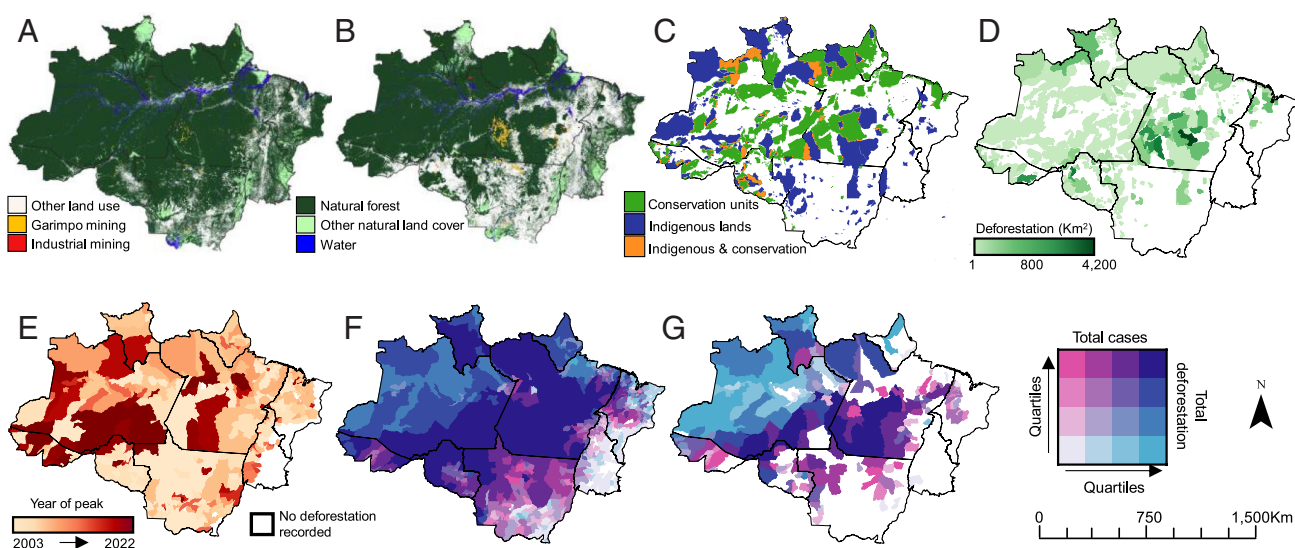


Fig. 1. Malaria and ecological change in the Brazilian Amazon. Land cover and land use in (A) 1985 and (B) 2022. (C) Officially demarcated conservation units and indigenous lands. (D) Total deforestation from 2003 to 2022 in conservation units and indigenous lands. (E) Year in which the peak of deforestation was observed in each municipality, 2003–2022. Bivariate maps of total deforestation, (F) total malaria cases, and (G) total malaria cases originating in indigenous localities, 2003–2022. All maps show state boundaries. State acronyms: RO: Rondônia, AC: Acre, AM: Amazonas, RR: Roraima, PA: Pará, AP: Amapá, TO: Tocantins, MA: Maranhão, and MT: Mato Grosso.

The state of Pará has four of the top five indigenous territories and three of the top five conservation units with the highest deforestation. Over time, the local peak of deforestation varied spatially. The peak of deforestation occurred in 2022 in northeastern Mato Grosso, northern Rondônia, and throughout Amazonas and Acre. In these areas, the more recent peak years often coincided with both conservation and indigenous protected areas (Fig. 1*E*). When considering cumulative values by municipality over the study period, it was observed that high deforestation and high malaria cases co-occurred along the south-central and eastern regions of the Amazon (Fig. 1*F*). This area is known as the arc of deforestation in Brazil (36). In addition, malaria cases in indigenous areas and deforestation co-occur most strongly along the arc of deforestation (Fig. 1*G*).

During the study period, 87.7% of all deforestation in the Amazon occurred in four states: Pará, Amazonas, Mato Grosso, and Rondônia. Additionally, Pará, Amazonas, and Rondônia accounted for 78.9% of all recorded malaria cases. We observed a significant correlation between malaria and deforestation for the entire Amazon ($0.62, P < 0.001$) and four states: Rondônia, Pará, Tocantins, and Maranhão (SI Appendix, Table S1). Scatterplots were generated to display the annual trajectories of deforestation and malaria cases from 1990 to 2022 (Fig. 2*A*). Deforestation and malaria cases were higher in 2022 than in 1990 only in Roraima

and Amazonas. The path to lower deforestation and malaria cases was complex, nonsimultaneous, and varied by state (Fig. 2*A* and *B*). The steepest decline in deforestation occurred during the first term of President Lula da Silva (2003 to 2011). Beginning with President Rousseff in 2011 and intensifying under Presidents Temer and Bolsonaro from 2017 to 2022, this trend has reversed (Fig. 2*C*). However, this annual relationship between malaria and deforestation only partially captures the complexity of frontier malaria, which occurs at fine spatial and temporal scales. To address this, we compiled monthly data on deforestation by municipality and malaria cases (municipality of infection) from January 2003 to December 2022. Monthly deforestation alerts were used in each municipality to apportion annual data by month (Materials and Methods). This is the most detailed and longest time series currently available to assess the impact of deforestation on malaria, while controlling for case importation, social, and environmental factors.

We found that the lag with the highest correlation coefficient was one month. This was supported by entomological and frontier malaria literature in the Amazon region and was included in the models (14, 17, 37–39). Total deforestation showed seasonality, with a peak in July across the entire Amazon, but varied by state (SI Appendix, Table S2). Roraima and Tocantins showed little evidence of seasonality (Fig. 3), while the seasonal peak of deforestation

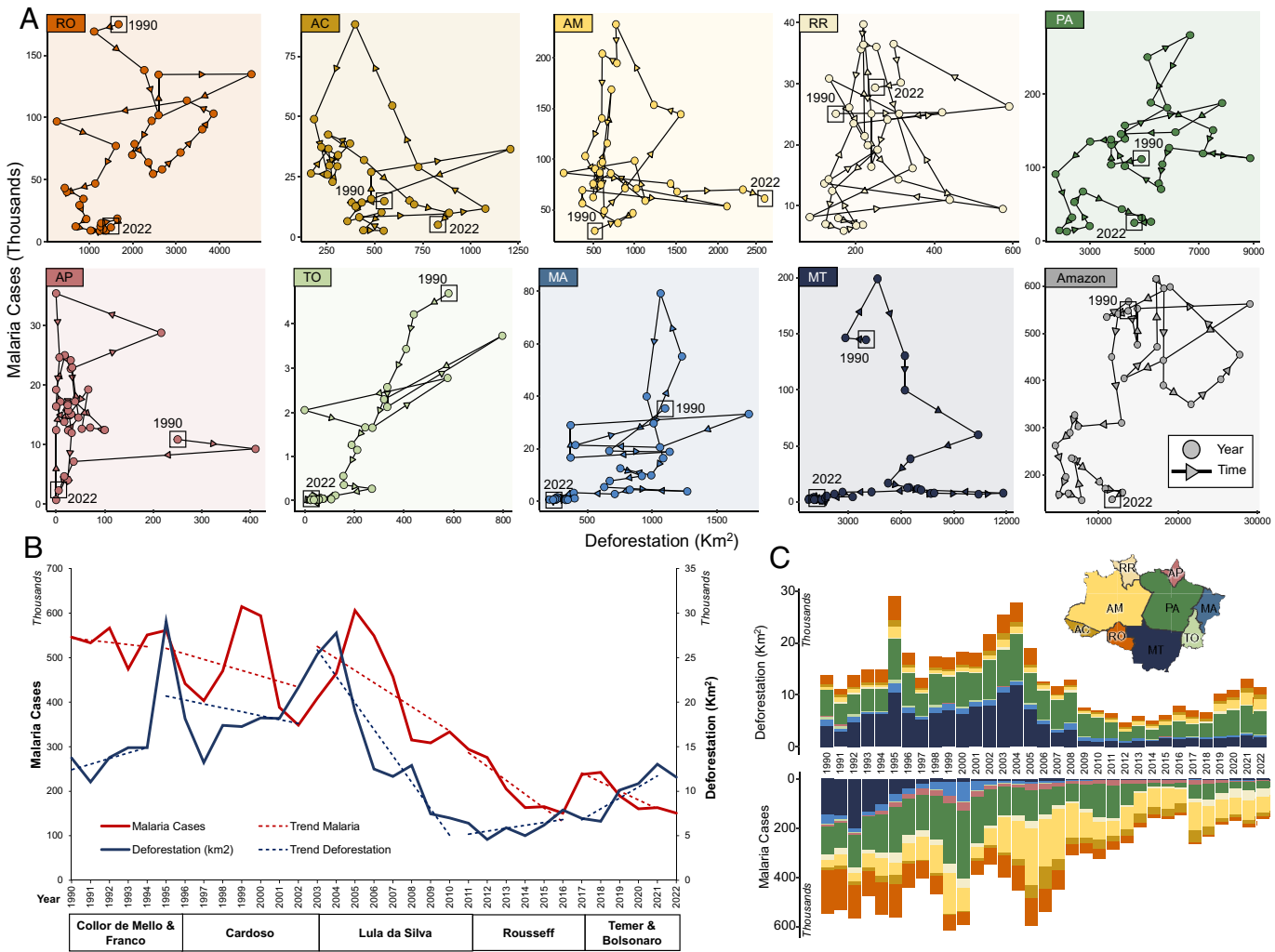


Fig. 2. Deforestation and malaria. (A) Temporal trajectory of annual deforestation (Km²) and malaria cases (thousands) by state and for the whole Amazon. Colors correspond to state colors shown in the map in panel C. (B) Annual malaria cases and deforestation rate in the Amazon, with best-fit lines specific to each political administration listed in the X-axis. (C) Annual deforestation rate and malaria cases color-coded by state. State acronyms: RO: Rondônia, AC: Acre, AM: Amazonas, RR: Roraima, PA: Pará, AP: Amapá, TO: Tocantins, MA: Maranhão, and MT: Mato Grosso.

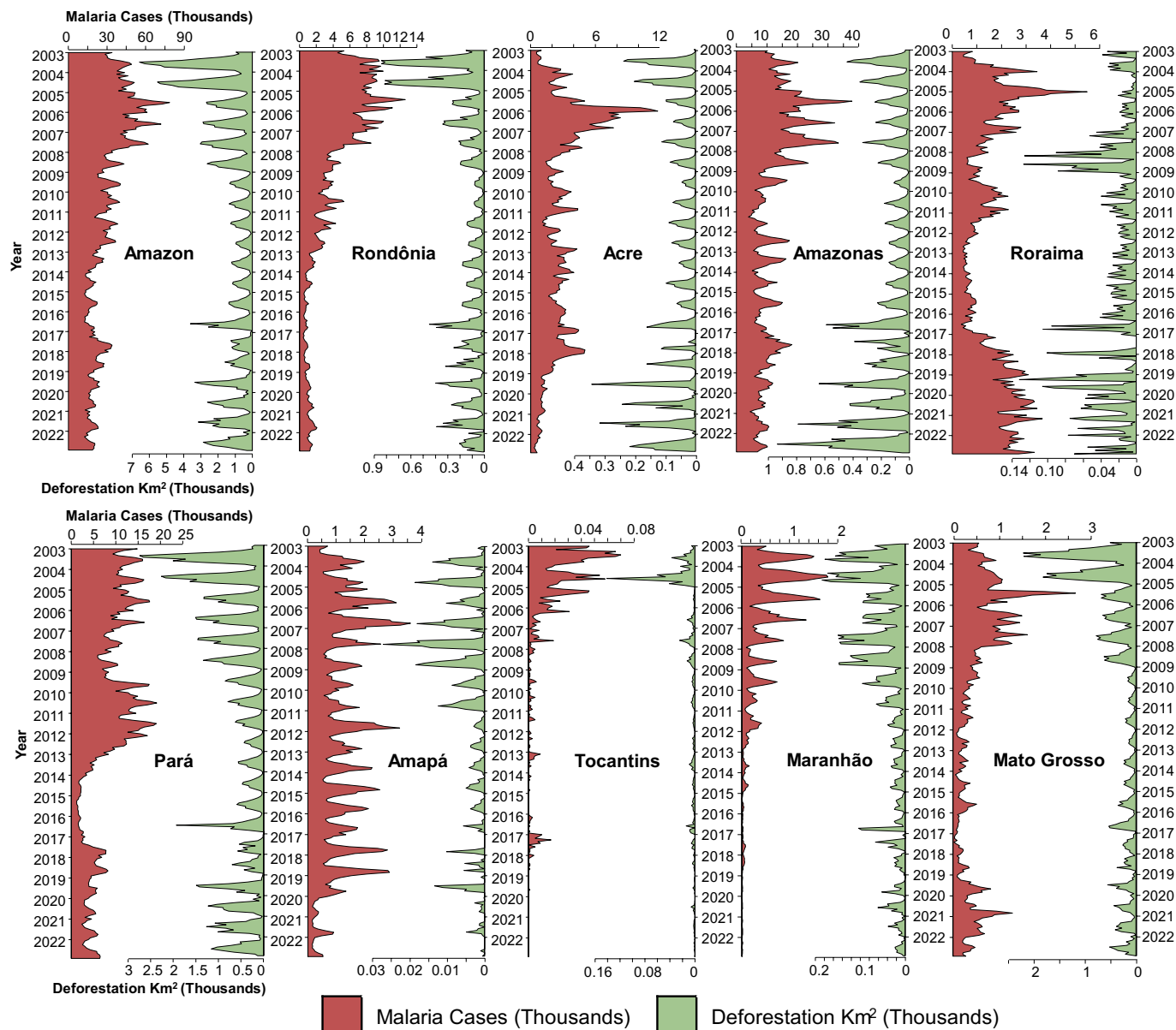


Fig. 3. Monthly trends in malaria cases and deforestation by state, 2003–2022.

occurred in August for Acre, Pará, and Amapá. Pará had the highest cumulative deforestation during the study period. Mato Grosso and Amazonas, which ranked second and third in terms of total deforestation, respectively, had seasonal peaks in July. Across the Amazon, the seasonal peaks of malaria transmission and deforestation differed by an average of 2.76 mo (SD: 2.42 mo), which varied by state. Amapá had the least variation in malaria and deforestation seasonality, with peaks differing by 0.90 mo (SD: 1.17 mo). On the other hand, Acre showed the most variation in seasonal peaks between deforestation and malaria cases, with a difference of 4.65 mo (SD: 2.18 mo) (*SI Appendix, Table S3*).

We expect the effect of deforestation on malaria to be lagged in time. According to a cross-correlation analysis, deforestation is most strongly linked to malaria, both overall and in each state, with a one-month lag (*SI Appendix, Fig. S1 A–E*). A Bayesian spatiotemporal hierarchical zero-inflated Poisson regression model was utilized to evaluate the correlation between the monthly deforestation rate (defined as the percentage of the municipality deforested with a 1-m lag) and malaria cases (Fig. 4 and *Materials and Methods* and *SI Appendix, Tables S4 and S5*). On average, a

1% increase in the monthly deforestation rate is followed by a 6.3% [95% credible interval (CrI): 6.2, 6.5] increase in malaria cases in the following month. We assessed the interaction between the 1-mo lagged monthly deforestation rate and the percentage of forest cover in a municipality. If a municipality has 1% more forest coverage, a 1% increase in deforestation is associated with a 24.58% (95% CrI: 23.7, 25.5%) increase in malaria cases in the following month. We found that monthly maximum temperatures between 27.69 to 29.59 and 31.50 to 34.68 °C were associated with more monthly malaria cases, while temperatures between 38.49 to 40.40 °C were associated with fewer monthly malaria cases. Additionally, La Niña and El Niño conditions were linked to an increase in malaria cases. Precipitation levels between 0 to 204 mm, 408 mm, 611 mm, and 713 to 866 mm were associated with an increase in monthly malaria cases, while levels between 917 to 1019 mm were associated with a decrease in cases (*SI Appendix, Fig. S2*).

We ran the same model stratified by state (Fig. 4 and *SI Appendix, Table S6*). The posterior distribution of the Tocantins model was too wide to plot. The effect of a 1% increase in the monthly deforestation

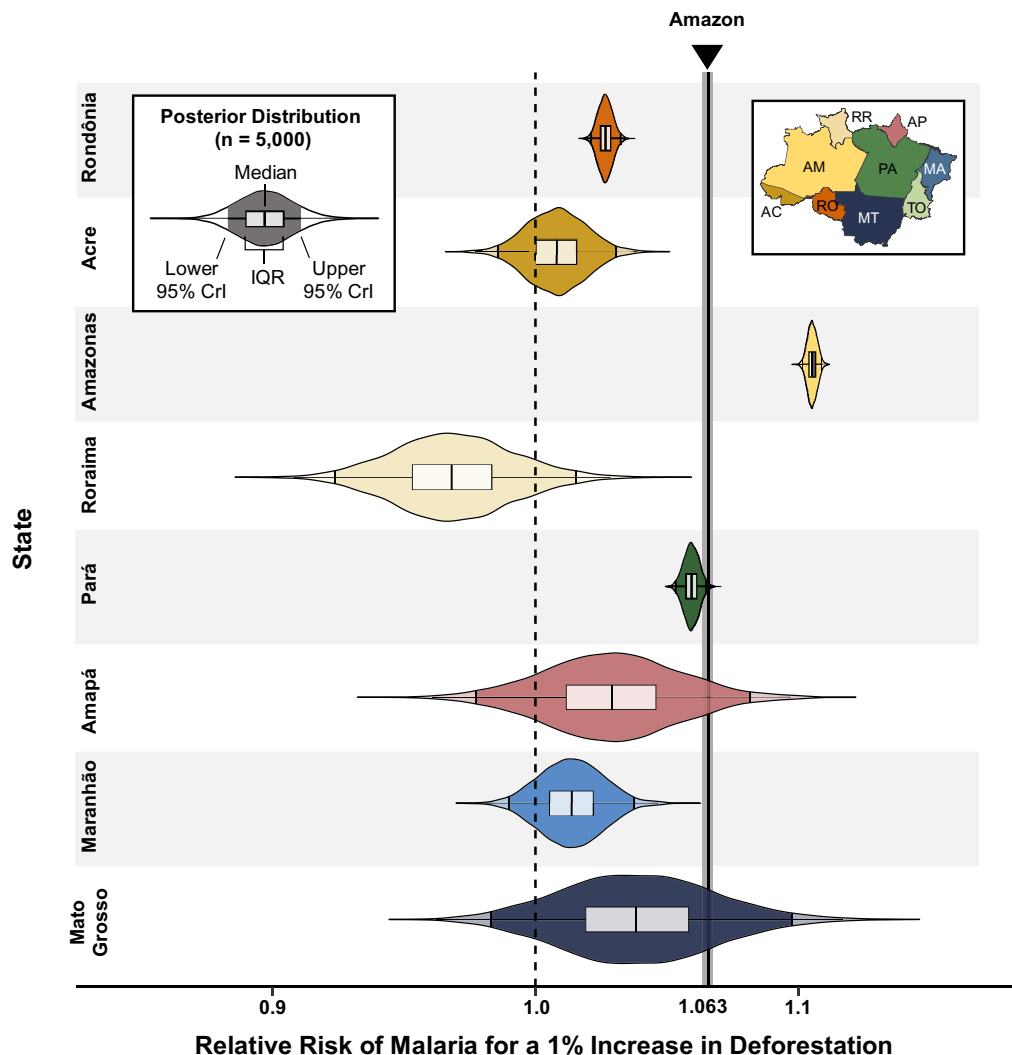


Fig. 4. Effect of deforestation on malaria. Posterior distribution (5,000 samples) from global and state-stratified Bayesian spatiotemporal models (*Materials and Methods*). Global model estimate and 95% credible interval shown by the black line and gray boundaries. In all models, the outcome is monthly malaria cases by municipality of infection. Colors correspond to states on the map.

rate on malaria varied by state and was statistically significant and positive in three states: 6.0% in Pará (95% CrI: 5.4, 6.6%), 10.6% in Amazonas (95% CrI: 10.2, 11.0%), 2.7% in Rondônia (95% CrI: 2.2, 3.3%). The interaction term between 1-mo lagged deforestation and the percentage of a municipality covered by forest was significant and positive in Pará, and significant and negative in Acre and Amazonas. The coefficient for mobility of malaria-infected individuals was highest in Tocantins. The coefficient for cases in indigenous localities was highest in Roraima and showed that a 1% increase in cases in indigenous localities was associated with a 62.5% (95% CrI: 51.6, 74.2%) higher number of autochthonous cases in the municipality. A similar trend was observed for cases in *garimpo* localities, such that in Roraima. A 1% rise in cases in *garimpo* localities corresponded to a 40.5% (95% CrI: 36.9, 44.3%) increase in autochthonous cases in that municipality.

Discussion

We used the longest and most spatiotemporally detailed data available to quantify the relationship between deforestation rate and malaria cases in the Brazilian Amazon. The findings indicate that deforestation rate is positively associated with malaria transmission in the proximal month across the entire region. This relationship

was observed in most states of the Brazilian Amazon, albeit to varying degrees. We found that the relationship between deforestation and malaria transmission was stronger in municipalities with a higher percentage of forested land area, and that the magnitude of this relationship varied by state. Furthermore, states that held the largest share of deforestation over the study period, Amazonas, Pará, and Rondônia, were the states with statistically significant relationships between malaria and monthly deforestation. Given the highly diverse ecological, sociodemographic, economic, and epidemiological contexts of the Brazilian Amazon, the underlying drivers of deforestation and subsequent malaria transmission must be carefully considered if this information is to be useful in Brazil's goal of malaria elimination.

In 2022, Brazil reaffirmed its commitment to *P. falciparum* elimination by 2030 and all malaria by 2035 (33). However, these efforts face significant challenges (40) due to two highly interdependent dynamics: environmental change and human mobility. To achieve elimination, it is crucial to understand these key factors (40). Our previous work has described the patterns of mobility, sources, and sinks of malaria-infected individuals across the Amazon. It showed that risk factors such as *garimpo* and forestry increase the likelihood of mobility (41). Here, we show that the ecological changes associated with these mobility-related activities,

when controlling for the effect of mobility itself, can be a driving force in increasing malaria transmission. This study emphasizes three crucial factors that can aid Brazil in achieving its goal of eliminating malaria.

First, Brazil is in a uniquely privileged position to enact data-driven intervention stratification policies for deforestation and malaria. Deforestation alerts are collected through Brazil's Deforestation Detection in Real Time (DETER) system (42), while the Malaria Epidemiological Surveillance Information System (SIVEP) collects near real-time, country-wide data on malaria cases and can track mobile cases (43). DETER categorizes alerts into underlying drivers of ecological degradation, such as selective cutting, mining, and fires, while SIVEP collects socioeconomic and demographic risk factors of malaria infection, such as occupation, age, and sex. Previous studies of the relationship between deforestation and malaria transmission have been conducted on an annual basis, and even the most comprehensive analyses have reported results for the entire Brazilian Amazon (29). We show state-level heterogeneity in the extent to which deforestation is associated with malaria transmission, even among the three states with the highest levels of deforestation. States such as Mato Grosso and Maranhão, which historically had high deforestation, declining malaria cases, and less forest cover, have a positive association between deforestation and malaria but a credible interval that includes the null. Considering the dynamics of frontier malaria, these states may be transitioning to a more human-dominated landscape, attenuating the effects of deforestation on malaria. Currently, the NMCP uses the annual API to stratify areas at high, medium, low, or no risk of malaria. Intervention strategies are tailored according to risk, regardless of local environmental dynamics (e.g., deforestation) (33).

Furthermore, annual associations mask the seasonal components of malaria transmission and deforestation, they lack information on the mechanistic time-lag between the two given frontier malaria dynamics and may bias the effect size of the association by not accounting for peak periods. We show that there are periods of year-to-year decline in both deforestation and malaria cases across the Amazon. McDonald & Mordecai demonstrate an annual, positive relationship between deforestation and malaria incidence that is higher in interior states of the Amazon with higher forest cover (29). However, these aggregated trends mask a monthly, stronger association between malaria transmission and deforestation by not capturing within-year deforestation maximums. Our paper additionally expands on the coarse spatial stratification used in McDonald & Mordecai and extends the analysis to the state-level. In the era of malaria elimination in the Brazilian Amazon, where transmission of *P. falciparum* is highly focal and often occurs in populations that are rapidly migrating for economic opportunity associated with deforestation, monthly, state-specific associations are more useful for guiding malaria control and elimination efforts (7).

According to the concept of frontier malaria, *P. falciparum* is usually the dominant parasite when forest is cleared (7). A major *P. falciparum* hotspot is located in western Amazonas, Acre (Juruá Valley), and Rondônia, and monthly deforestation was found to be positively associated with malaria in all three states. In the Juruá Valley, Brazil's most persistent malaria hotspot (44), the relationship between malaria, deforestation, and mobility is unique. In contrast to malaria dominating rural areas in other parts of the Amazon, periurban and urban areas in this region experience a higher incidence of malaria infections due to mobility and infections associated with seasonal farming (40, 44, 45). Mobile populations may contribute to the spread of malaria to neighboring regions, leading to deforestation

effects beyond the main area affected by the forest removal (e.g., periurban and urban settings). The largest source of malaria infections in the Juruá Valley is Guajará in Amazonas, and the most common route of travel from there is to Cruzeiro do Sul in Acre (41). Guajará's peak deforestation occurred in 2022, while deforestation in other parts of the Juruá Valley peaked in earlier years, indicating an expansion of ecological change. As connectivity increases between urban centers and newly deforested land in this region due to extended urbanization, malaria elimination efforts may face even higher transmission rates. Therefore, controlling for mobility is an essential component of our analysis as it allows us to accurately measure the impact of deforestation on malaria transmission without the confounding effects of human movement. Surveillance, as an intervention, is a core pillar of the WHO Global Technical Strategy for Malaria 2016–2030 (46). To achieve the goal of eliminating *P. falciparum* in Brazil, it may be useful to utilize real-time deforestation warnings from DETER to monitor the opening of settlements/agricultural land in the Juruá Valley and beyond. Additionally, tracking mobility, occupation, and parasite species of infected individuals to/from these regions using SIVEP data can aid in allocating interventions that are tailored to the complex local contexts.

Second, understanding the subannual effects of deforestation on malaria provides insight into the response of malaria vectors to ecological changes. This study confirms the dynamics of frontier malaria, specifically the synergistic effects of forest cover and deforestation on malaria transmission (14). The proliferation of vector habitats, higher host density, and frequent human-vector contact combine to create this increase. There is evidence that areas with higher rates of deforestation are associated with higher *Ny. darlingi* density (10, 25). Additionally, deforestation in heavily forested areas has been found to result in a significant increase in malaria cases in the following month. Moreover, under favorable climatic conditions, *Nyssorhynchus* can complete its life cycle in just 10 to 14 d. Thus, it is likely that if the vector is already present in the deforested area, it will proliferate easily within one month after deforestation, provided that precipitation creates adequate habitats. Our models control for weather- and climate-related variables, which are vital components of vector development. ONI values beyond the thresholds for La Niña/El Niño were associated with heightened risk of malaria infection, as were cooler temperatures which may be associated with rainier months. However, temperature, precipitation, and extreme weather patterns vary across the Brazilian Amazon (47), and thus vector response to those factors is likely inconsistent across the region. Vector control strategies in the Brazilian Amazon, particularly in new settlement and mining communities, present significant challenges. Indoor residual spraying is ineffective when houses are highly porous, and when biting mainly takes place outdoors. Insecticide-treated net coverage and uptake are low, and this measure can be of limited impact if the vector mainly bites outdoors. Larviciding has been viewed as impracticable from a logistical standpoint (40), but can be highly effective in specific settings in the Amazon, such as fishponds (48). Above all, our findings contribute to understanding the monthly relationship between deforestation and malaria transmission at the local level, which supports better vector control planning. The best combination of vectors' control strategies will, however, depend on the local context.

Third, malaria is increasing in vulnerable populations, particularly indigenous communities and *garimpo* workers, and ecological change is at the heart of these increases (49). Although the first Lula da Silva administration implemented policies to conserve

forests and protect indigenous territories (50), subsequent political administrations have dismantled these programs (51, 52). During the Temer and Bolsonaro administrations, illegal land grabbing and *garimpo* activities expanded uncontrollably, even in indigenous territories and conservation units. This expansion has led to a dramatic increase in malaria cases in these regions (23, 52). The Yanomami people are of particular concern, as the newly re-elected Lula da Silva government has declared a public health emergency, citing *garimpo* activities as the main cause of an increase in malaria, sexually transmitted infections, and mercury poisoning (22). The Yanomami territory includes parts of northern Roraima and Amazonas, but most of the *garimpo* activity occurs in Roraima. Although highly invasive and acutely harmful to health, *garimpo* causes a relatively small proportion of deforestation compared to agriculture and other drivers (34). We found that the deforestation coefficient in Roraima is negative and not statistically significant, but that the coefficients for *garimpo* cases and indigenous cases are surprisingly high. On the one hand, this may be an example of small-scale deforestation due to mining being particularly harmful for malaria transmission. On the other hand, Roraima experiences rapid cross-border movement between neighboring countries and states for mining and other reasons (53), which may contribute to local transmission in Roraima despite ecological changes occurring elsewhere. Although the active *garimpo* frontier is constantly moving, this type of mining leaves behind a trail of abandoned mines that contain dense larval habitats. These habitats can lead to sustained high levels of malaria transmission in nearby, stationary communities (54, 55). Additionally, these abandoned mines have also been linked to mercury exposure in surrounding populations (56). Although *garimpo* mining is being curtailed by new policies, habitats that support ongoing transmission will remain and can serve as areas for increased surveillance. However, there are other challenges that compound this dynamic. Knowledge of disease transmission among miners is limited, sites are remote, housing conditions are poor, and populations are highly mobile (57). The concentration of these malaria hotspots makes interventions such as Malakit (58) more feasible. Malakit allows mobile populations to diagnose and treat themselves without having to travel to a malaria outpost. Interventions such as Malakit, when combined with robust active surveillance, can help reduce parasite burden in mobile populations. However, reinvigorated political will and enforcement of conservation and indigenous territory laws will be critical to reducing the spread of malaria to nearby at-risk communities.

This study has several strengths. First and foremost, the dataset we use considers a spatiotemporal scale that has not been previously explored, which allows for a more nuanced comprehension of the ecological and epidemiological mechanisms that underlie the relationship between deforestation and malaria, considering known dynamics such as frontier malaria. Also, unlike other studies, we have controlled for various key factors that could potentially challenge malaria elimination in Brazil. This has allowed us to isolate the unique effect of deforestation without confounding influences. To date, no study has considered the dynamics of human mobility in the Brazilian Amazon when assessing the relationship between deforestation and malaria transmission.

This study has limitations. Monthly deforestation values were calculated from alert data, rather than derived directly from satellite imagery. However, these data represent the best proxy for subannual intensity of deforestation activity available. Additionally, the analysis was conducted at the municipality level. Municipalities in the Brazilian Amazon vary greatly in size. Therefore, in larger municipalities, deforestation may not be precisely located in areas with the highest burden of transmission. However, finer spatial

scales are not available. Additionally, our primary exposure, the monthly deforestation rate, does not account for the subsequent land tenure category to which forested land is transitioned. Deforestation for large-scale agribusiness may have a different impact on malaria dynamics compared to smaller-scale clearing for mining or settlement. This is because it may result in fewer people encountering deforested areas. However, if this is the case, it could bias coefficients on deforestation toward the null, thus underestimating its effect on malaria transmission.

Conclusion

Brazil faces challenges in its efforts to eliminate malaria. While cases are declining, transmission in indigenous and *garimpo* areas is increasing significantly due to ecological changes. These changes have extended into previously protected areas, posing unique challenges for the Brazilian government. In this study, we show a clear positive association between deforestation and malaria transmission in the short term. Our findings provide a critical foundation for quantifying the impact of ecological changes on malaria risk on a monthly basis. Brazil can use its rich surveillance data on malaria cases and ecological changes to incorporate new intervention stratification methods into its plan to eliminate malaria by 2035, but also to respond quickly to environmental violations.

Materials and Methods

Study Area. Brazil is the third-largest country in the Americas, covering an area of 8,510,345 km². It consists of 26 states and one federal district, which are further divided into 5,570 municipalities. The Brazilian Amazon spans over nine states and 756 municipalities and accounts for 99.9% of the reported malaria cases in the country. Malaria diagnosis and treatment in Brazil are provided free of charge by the Brazilian government through an extensive network of laboratories and health centers.

Data Sources. We obtained malaria case data from the Malaria Epidemiological Surveillance Information System (SIVEP), which is supported by the Ministry of Health and stores all malaria test results. SIVEP data are of high quality, and low response variables are not those of interest for our study (59). Cases are confirmed either by blood smear microscopy (about 96% of cases) or by a rapid diagnostic test (RDT). We use data on reported positive and negative malaria tests from 2003 to 2022, aggregated by municipality and month. We also considered importation status. In Brazil, a locally acquired case is defined as a case in which the notification municipality is the same as the municipality of infection, regardless of residence location, while an imported case is defined as a case in which the notification municipality is different from the municipality of infection, regardless of residence location. SIVEP contains data that allow the assessment of importation status (43). Finally, we extracted data on the type of locality in which the case originated. Localities are submunicipal areas used by the NMCP. Unfortunately, digital shapefiles of localities are not available, which prevents spatial analysis at this scale. Localities are categorized as i) indigenous—occurring in indigenous lands, ii) rural, iii) urban, iv) *garimpo*—occurring in areas with artisanal mining, and v) settlement—occurring in areas with an established agricultural settlement.

Deforestation data for each municipality were assembled using two monitoring systems from the Brazilian National Institute for Space Research (INPE) (*SI Appendix, Fig. S3*). PRODES (Measurement of Deforestation by Remote Sensing) provides annual data for deforestation since 1988 (60). DETER (Deforestation Detection in Real Time) is a remote-sensing-based alert system for forest degradation with daily resolution extending from July 2016 to near-real time (42). To estimate municipal deforestation rate by month, we apportioned PRODES data based on the monthly distribution of alerts' area captured by DETER. Specifically, we calculated the total area of deforestation alerts for each municipality in each month for the period July 2016 to December 2022. Then, we converted this to a percentage based on the total area of alerts in each year and municipality, now referred to as alert intensity distribution. We multiplied this alert intensity distribution based on the true DETER alert data to PRODES annual data between

2016 and 2022. For years that did not have DETER data (December 2003 to June 2016) we took the average alert intensity distribution in each municipality for months in July 2016 to December 2022. This average alert intensity distribution was then applied to all other years in the dataset and for each municipality. Since random variation is expected to occur month-to-month within years, we incorporated a conservative random term into these monthly metrics based on the municipality's alert intensity distribution coefficient of variation for the years 2016–2022 (Mean/SD). Specifically, we created a normal distribution for each municipality with mean zero and SD equal to $1/100^{\text{th}}$ the municipality and month's coefficient of variation. We added those values specific to each municipality/month combination to all 17 interpolated years. Incorporating this random variation is a simple heuristic technique to introduce variability based on the true statistical characteristics of the dataset, while maintaining the lower bound of the deforestation data at zero. To assess validity of this interpolated dataset, we took the municipality average of the monthly deforestation values from the true alert intensity distributions for 2017 to 2022 and the average of the interpolated monthly values using the average alert intensity distribution, including random variation, for this same period across all municipalities in the Amazon within each year and visually assessed trends (SI Appendix, Fig. S4). We also looked at the annual rate of deforestation compared to the year prior. We calculated this variable for each municipality by subtracting the previous year's total deforestation from the current year, and then we divided that value by the municipalities' total area.

Daily precipitation and maximum temperature data were extracted from ERA5, the fifth global reanalysis product of the European Center for Medium-Range Weather Forecasts (61). ERA5 is a reanalysis dataset that combines meteorological observations with complex atmospheric modeling to produce a global grid of data. It is available at hourly temporal resolution and a 30 km spatial resolution. However, we use the daily aggregated version. We accessed ERA5 data using Google Earth Engine, a geospatial computing platform where daily ERA5 data are available. Because many municipalities are larger than the 30 km grid cell of the ERA5 product, we aggregated meteorological variables into a single daily value by using the precipitation and temperature observations at the most populated point in each municipality (62). For gridded population data, we used the WorldPop dataset (63). All weather variables were aggregated on a monthly scale. Oceanic Niño Index (ONI) data were extracted from the National Oceanic and Atmospheric Administration between 2003 and 2022 (64, 65). ONI is a 3-m rolling average of the difference from the mean sea surface temperature in the waters of the east-central tropical Pacific Ocean. Index values of +0.5 or higher indicate El Niño conditions, while values of −0.5 or lower indicate La Niña conditions. The final month in the 3-mo rolling average was merged into the case data and considered as the exposure for that month.

We obtained the road network by type (paved, unpaved) in kilometers, and land cover/use data for the years 2003–2022 from the Brazilian Annual Land Use and Land Cover Mapping Project (MapBiomas) (34). MapBiomas provides aggregated statistics of land cover/use data by year and municipality. From these datasets, we extracted the total area covered by natural vegetation each year. In addition, the area covered by conservation units and indigenous territories was calculated using shapefiles provided by INPE. Additionally, data on the total population by year and the area of the municipality were obtained from the Brazilian Institute of Geography and Statistics (IBGE) (66, 67).

Seasonality and Lag Periods. To assess the seasonality and temporal lag patterns between deforestation and malaria, we utilized a multiplicative decomposition, correlograms, and a cross-correlation function (SI Appendix, Fig. S5). We performed autocorrelation analysis for temperature, precipitation, and ONI to determine suitable nonlinear modeling structures (SI Appendix, Fig. S1).

Spatiotemporal Modeling. We used a Bayesian hierarchical zero-inflated Poisson model with spatial, temporal, and spatiotemporal covariance structures to quantify the effect of deforestation on malaria cases (68, 69), controlling for selected variables (SI Appendix, Table S4). We use malaria cases as our outcome with population as a covariate for two reasons: 1) the outcome of malaria cases is more interpretable than cases per unit population, and 2) we anticipate population to confound the relationship between deforestation and malaria transmission. Control variables were selected based on scientific knowledge of their association with malaria transmission in the Amazon (14, 70–73). The outcome variable (y_{jptk})

was monthly malaria cases in each municipality j , state p , month t , and year k , assumed to follow a zero-inflated Poisson distribution. The model had the form:

$$\begin{aligned} \log(y_{jptk}) = & \beta_0 + \beta_1 (\text{Deforestation rate}_{jt}) \\ & + \beta_2 (\text{Annual deforestation change}_{jk}) \\ & + \beta_3 (\text{Ratio imported to local}_{jt}) + \beta_4 (\text{Forested area}_{jk}) \\ & + \beta_5 (\text{Forested area}_{jk} * \text{Deforestation rate}_{jt}) + \beta_6 (\text{Year}) \\ & + \beta_7 X_{jt} + \beta_8 X_{jk} + \beta_9 X_j + f(\text{Temperature}_{jt}) \\ & + f(\text{Precipitation}_{jt}) + f(\text{ONI}) + f(\text{Month}) \\ & + q_j + v_p + u_j + \varepsilon_{jptk}, \end{aligned}$$

$$q_j \sim N(0, \sigma_q^2), v_p \sim N(0, \sigma_v^2), u_j \sim N(0, \sigma_u^2), \varepsilon_{ijtk} \sim N(0, \sigma_\varepsilon^2),$$

where β_0 is the intercept, $\beta_1(\text{Deforestation rate}_{jt})$ is the coefficient for the 1-mo lagged percentage of forest loss in the municipality of infection j and month t , $\beta_2(\text{Annual deforestation change}_{jk})$ is the coefficient for the difference in forest loss between years k and $k-1$ in municipality j corrected by the area in municipality j , $\beta_3(\text{Ratio imported: local}_{jt})$ is the coefficient for the ratio of imported cases to locally acquired cases in the municipality of infection j and month t , $\beta_4(\text{Forested area}_{jk})$ is the percentage of the municipality covered by forest in the municipality of infection j in year k , $\beta_5(\text{Forested area}_{jk} * \text{Deforestation rate}_{jt})$ is the interaction term between deforestation rate and forested area, $\beta_6(\text{Year})$ is a vector of coefficients for year fixed effects, $\beta_7 X_{jt}$ is a vector of coefficients for covariates observed in municipality j and month t (percentage of malaria cases occurring in *garimpo*, indigenous, settlement, and rural locality types), $\beta_8 X_{jk}$ is a vector of coefficients for covariates observed in municipality j and year k (kilometers of paved and unpaved roads, and total population); and $\beta_9 X_j$ is a vector of coefficients for covariates observed in municipality j (municipality area, percentage of the municipality covered by conservation units in 2022). Monthly maximum temperature $f(\text{Temperature}_{jt})$, monthly total precipitation $f(\text{Precipitation}_{jt})$, and the Oceanic Niño Index $f(\text{ONI})$, were modeled as first-order autoregressive processes. Month of year $f(\text{Month})$, was modeled as a cosine function optimized by residual sum of squares optimization for the whole Amazon (and for each state in stratified models) to account for temporal dependencies in malaria transmission. The term q_j is the municipality-level random effects component, v_p is the state-level random effects component, u_j is the spatial autocorrelation structure between municipalities (modeled as Besag-York-Mollié), and ε_{ijtk} is the municipality-month of year interaction term accounting for residual variation. We mapped the posterior mean random effects to assess for correlation across space (74) (SI Appendix, Fig. S6). Weakly informative priors were assumed for both regression parameters and hyperparameters. Estimation of the posterior marginal distributions was performed using the Integrated Nested Laplace Approximations (INLA) after setting the prior distribution for the model parameters (the intercept, coefficients, spatial and temporal effects, and residual effects) (75).

To account for potential state idiosyncrasies, we run the same model stratified by each state in the legal Brazilian Amazon. The variables used in the stratified models were identical to those in the model described above, except for the state-level random u_j . All statistical analyses were performed in R version 4.1.3 (76). All mapping was done in ArcGIS Pro and ArcMap 10.8.2 (77).

Sensitivity Analyses. Model robustness checks were conducted to evaluate different choices of penalties for the semiparametric trends used, specifically a time trend for sequential year-months and splines for temperature/precipitation. In addition, we tested different model specifications based on the removal of municipality-level random effects, the inclusion of state-level fixed effects, and the removal of year-level fixed effects (SI Appendix, Tables S7 and S8). Model goodness of fit was assessed using leave-one-out (LOO) cross-validation via comparison of the summed conditional probability of each observed data point (CPO), DIC comparison, marginal likelihood comparison, and WAIC comparison, with particular weight given to the LOO estimates because of their increased validity reported in the literature (78) (SI Appendix, Table S9). We also compare final model goodness

of fit based on four distributional assumptions: Poisson, zero-inflated Poisson, negative binomial, and zero-inflated negative binomial (SI Appendix, Table S10).

Data, Materials, and Software Availability. All data to recreate the analyses in this manuscript, along with the R scripts, can be found at: <https://github.com/mcastrolab/Deforestation-Malaria-Brazil> (79). The data grouped and merged to create the dataset placed in the GitHub were all publicly available (outlined in the manuscript text) except data on the notifications that tested negative for malaria, which were obtained through special request from the Brazilian Ministry of Health. Parties interested in obtaining such data can solicit a request via the Ministry of Health. Deforestation data are available at <https://terraberil.dpi.inpe.br/en/download-files/> (42, 60). Weather data are available at <https://cds.climate.copernicus.eu/datasets> (61). Shapefiles for Brazil are available at <https://www.ibge.gov.br/en/geosciences/territorial-organization/territorial-meshes/18890-municipal-mesh.html> (66). Land cover data for road surfaces and conservation

area are available at <https://brasil.mapbiomas.org/en/colecoes-mapbiomas/> (34). Population data are available at both www.worldpop.org (63) and <https://www.ibge.gov.br/en/statistics/social/population/18448-estimates-of-resident-population-for-municipalities-and-federation-units.html> (67).

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